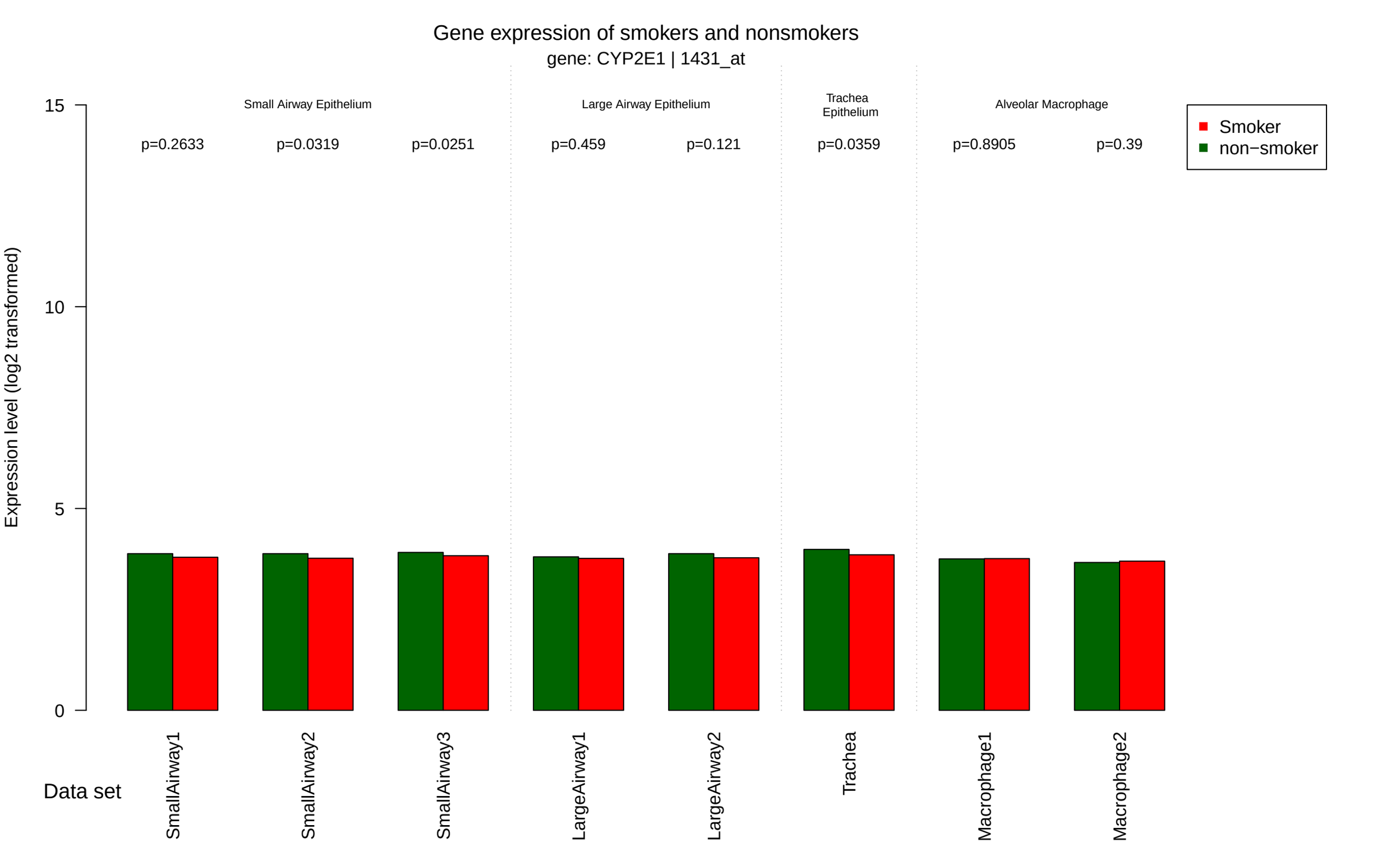
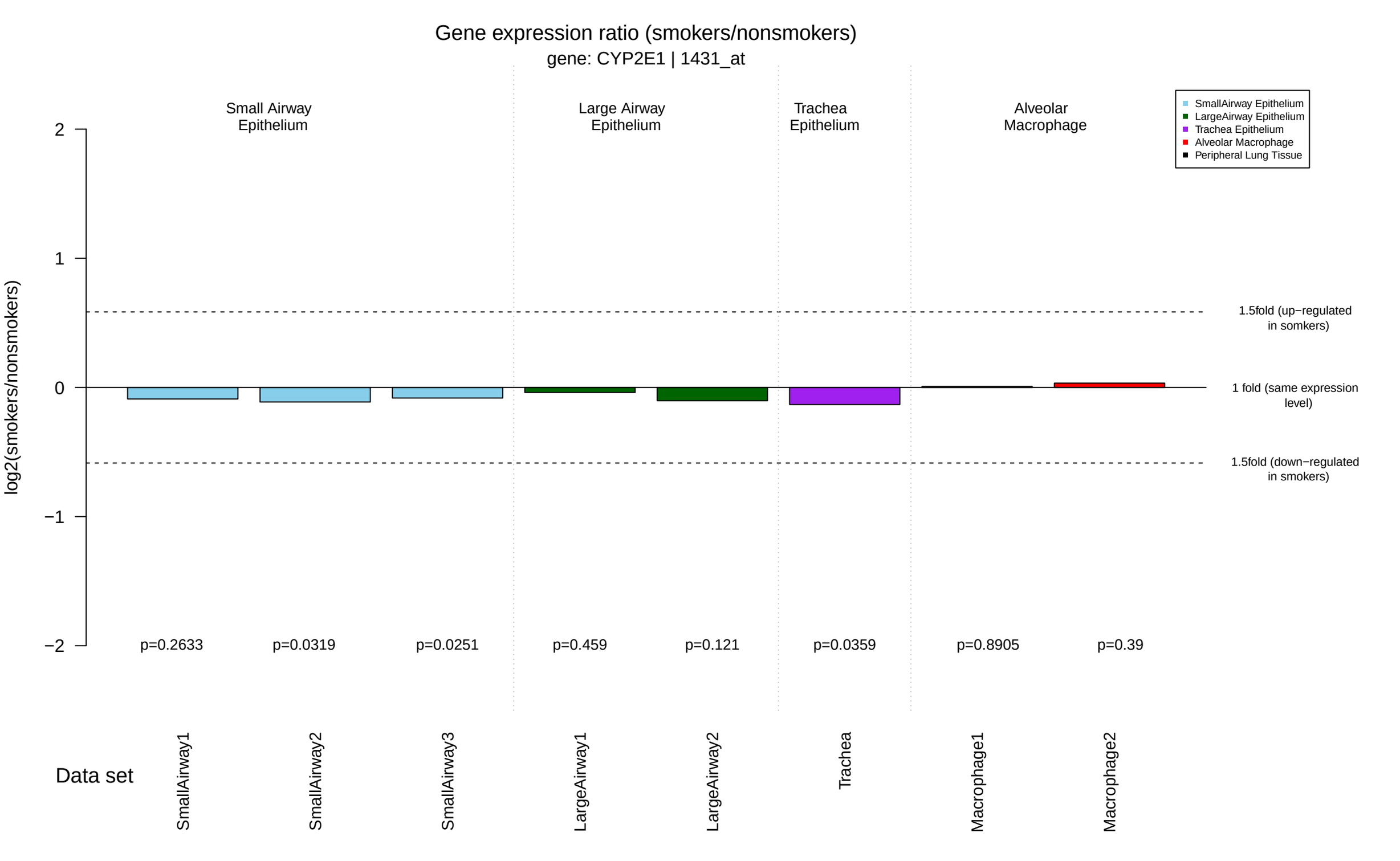
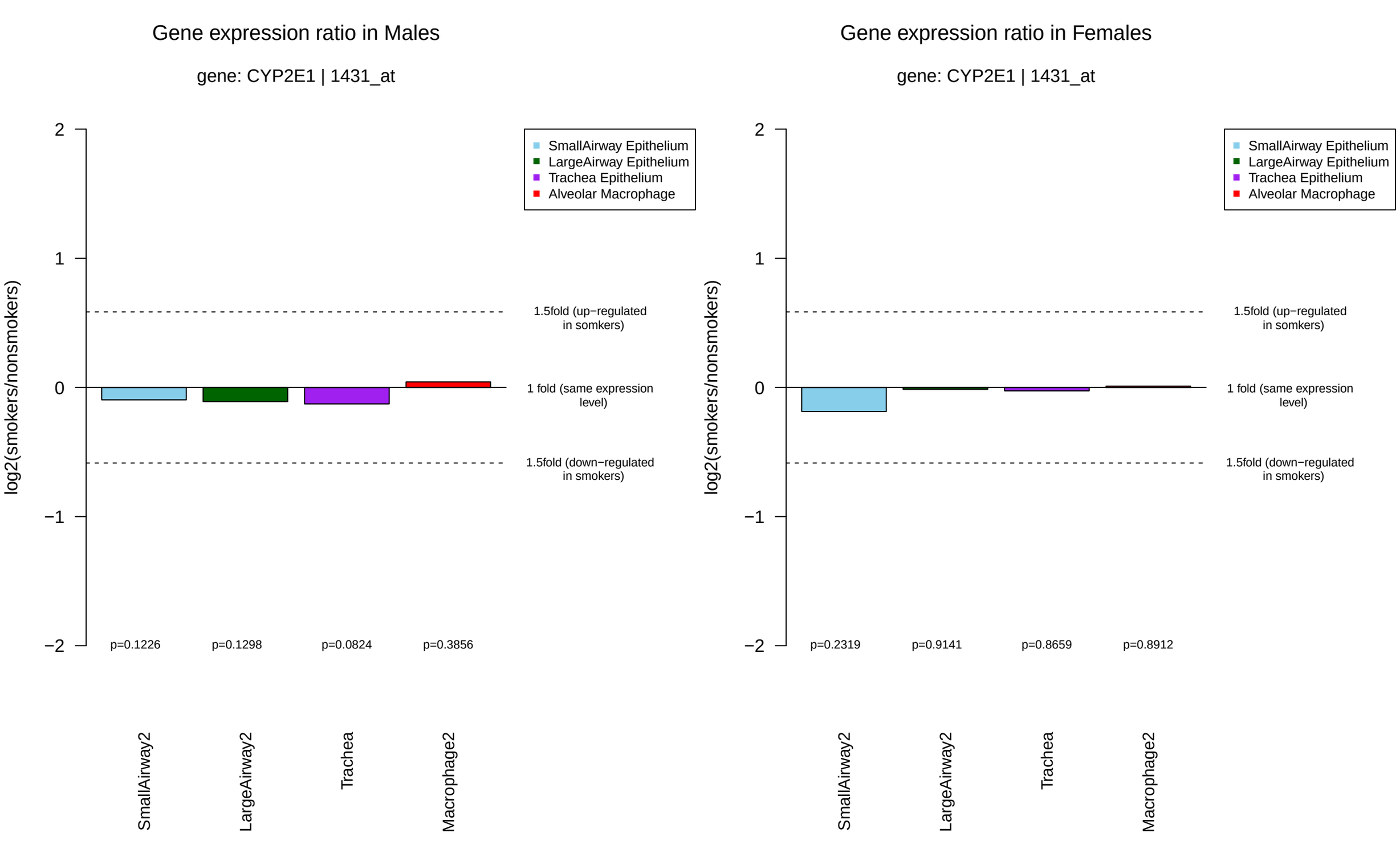
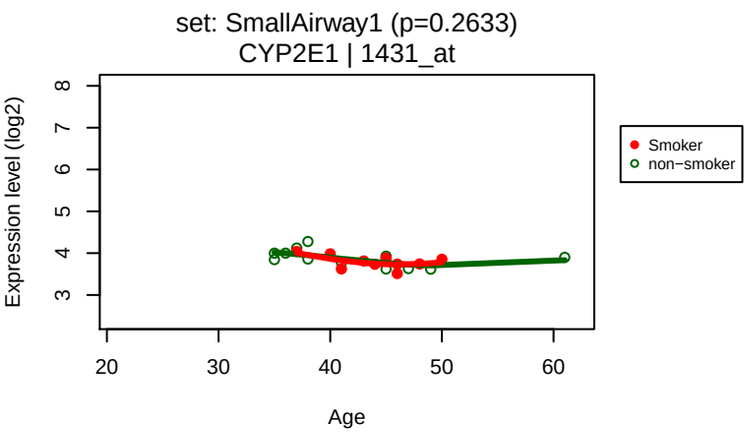
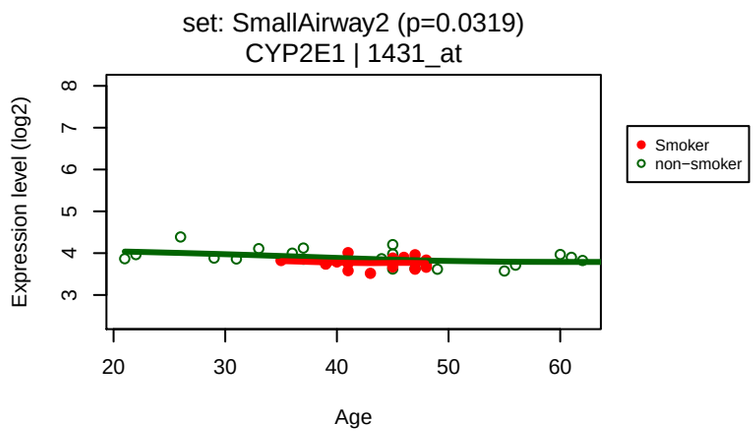
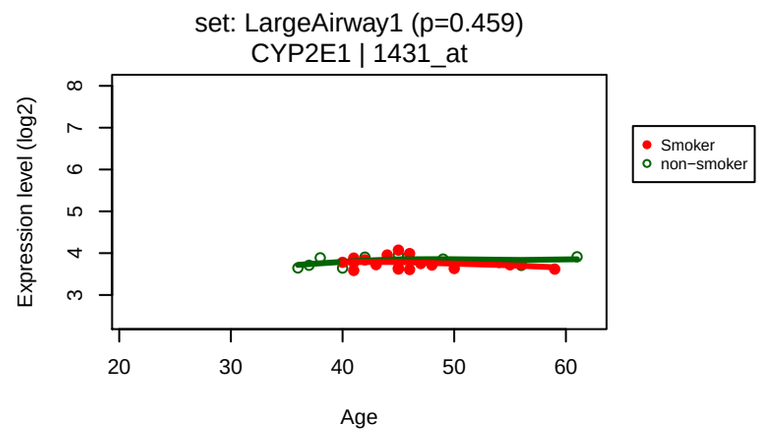

Gene expression of smokers and nonsmokers
gene: CYP2E1 | 1431_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2633
p=0.0319
p=0.0251
p=0.459
p=0.121
p=0.0359
p=0.8905
p=0.39
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

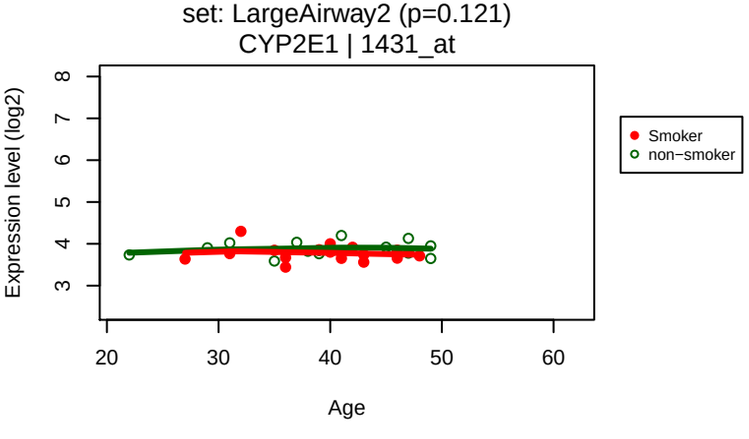

Gene expression ratio (smokers/nonsmokers)
gene: CYP2E1 | 1431_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2633
p=0.0319
p=0.0251
p=0.459
p=0.121
p=0.0359
p=0.8905
p=0.39
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

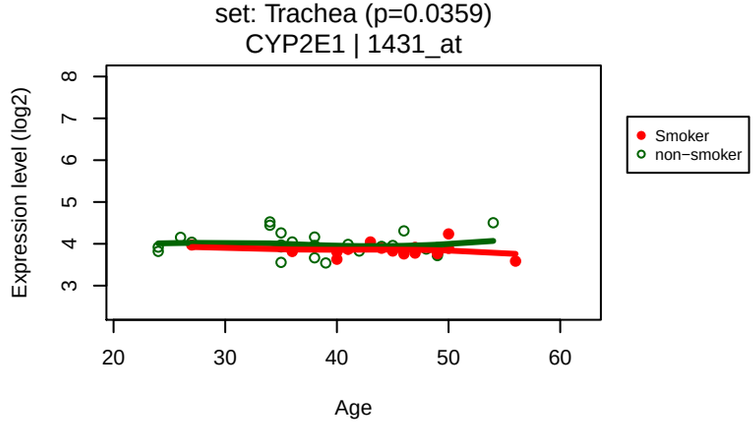

Gene expression ratio in Males
gene: CYP2E1 | 1431_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1226
p=0.1298
p=0.0824
p=0.3856
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP2E1 | 1431_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2319
p=0.9141
p=0.8659
p=0.8912
SmallAirway2
LargeAirway2
Trachea
Macrophage2

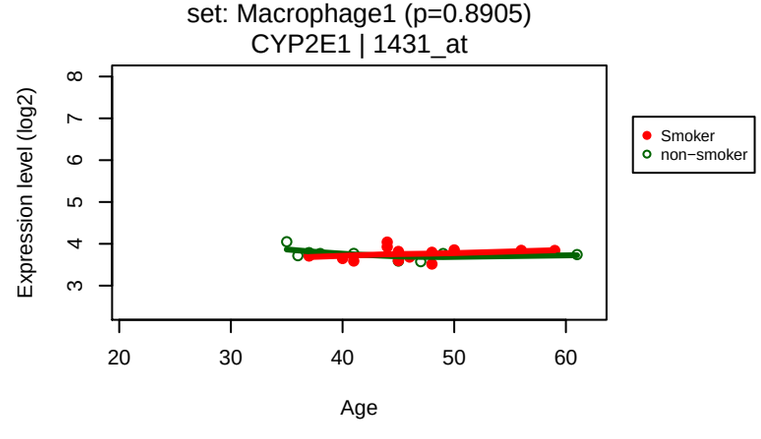

set: SmallAirway1 (p=0.2633)
CYP2E1 | 1431_at
Expression level (log2)
Age
Smoker
non−smoker

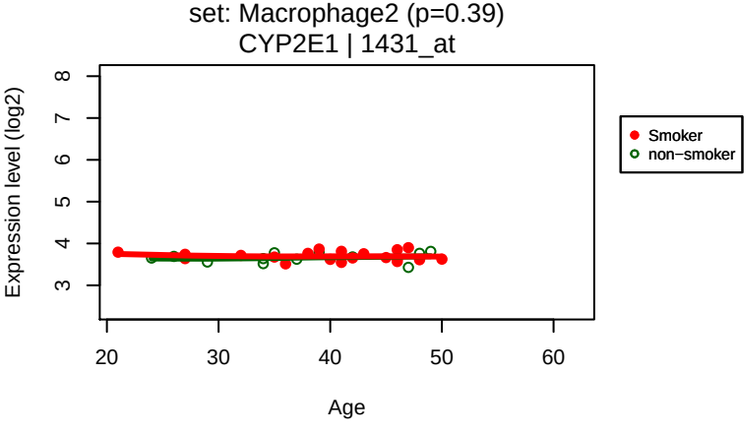

set: SmallAirway2 (p=0.0319)
CYP2E1 | 1431_at
Expression level (log2)
Age
Smoker
non−smoker

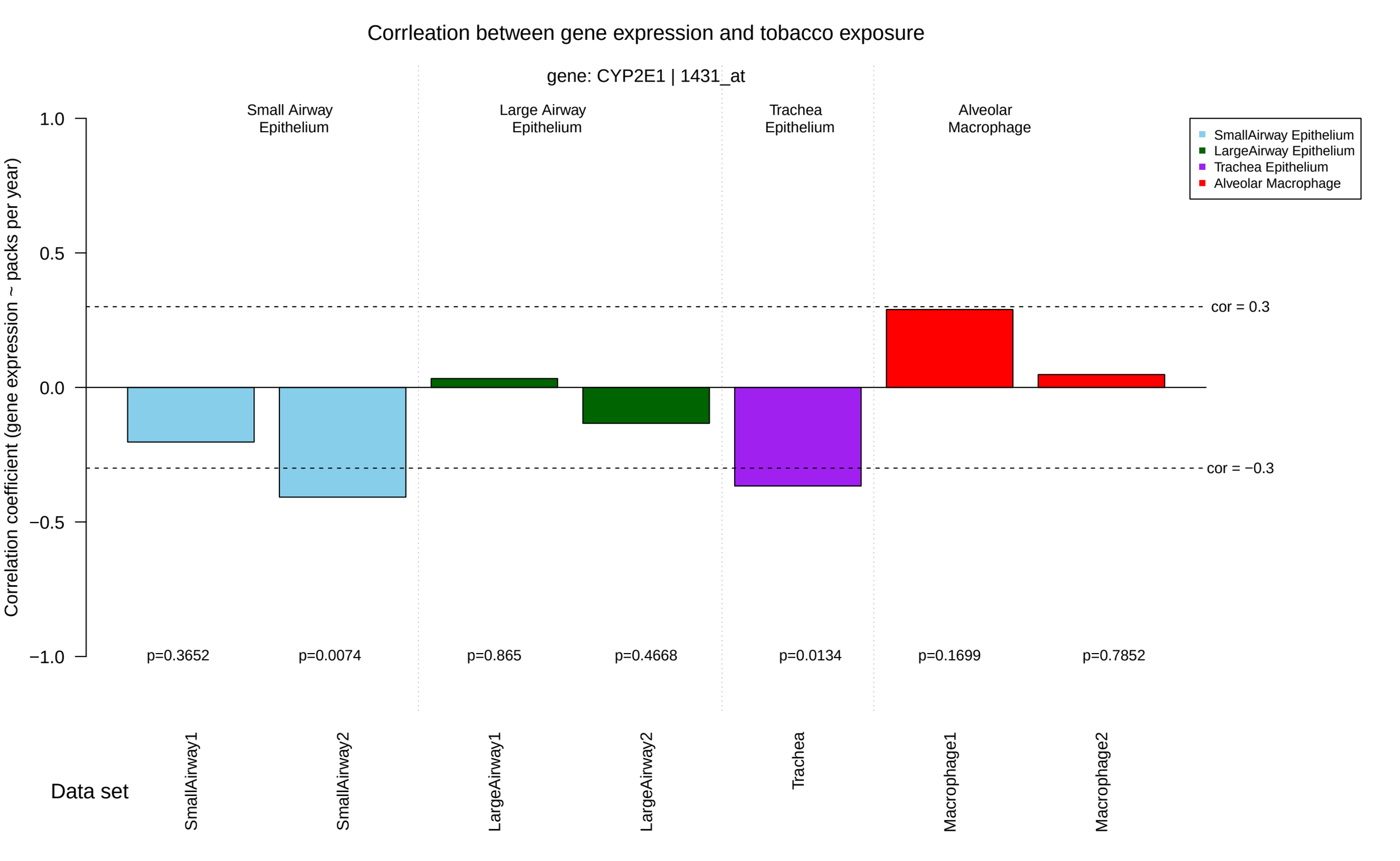

set: LargeAirway1 (p=0.459)
CYP2E1 | 1431_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.121)
CYP2E1 | 1431_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0359)
CYP2E1 | 1431_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8905)
CYP2E1 | 1431_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.39)
CYP2E1 | 1431_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP2E1 | 1431_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3652
p=0.0074
p=0.865
p=0.4668
p=0.0134
p=0.1699
p=0.7852
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2